

Predictors of Opioid Misuse Disorder in the US

Final Project

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1 Abstract

Background: Opioid use disorder and pain reliever misuse remain major public health concerns in the United States.

Research Question: What state-level factors are most useful for predicting pain reliever misuse?

Methods: We used state-level National Survey on Drug Use and Health data from 2018–2019 and 2021–2024. After exploratory data analysis and Box-Cox transformation of numeric predictors, we compared a full linear regression model, a stepwise AIC model, and a LASSO model. Model selection was based on 10-fold cross-validated RMSE.

Results: The final LASSO model identified educational attainment, poverty, age composition, employment, Medicaid coverage, and racial composition as the strongest predictors of pain reliever misuse. Some college education, the share of children aged 12 to 17, full-time employment, and Medicaid coverage were positively associated with misuse, while higher shares of people with a college degree, less than a high school education, incomes above two times the federal poverty level, and non-Hispanic white and Black residents were negatively associated with misuse.

Conclusion: The strongest state-level predictors of pain reliever misuse were educational attainment, poverty, and age composition, with additional contributions from employment, Medicaid coverage, and racial composition. Because the data are cross-sectional and aggregated at the state level, the results should be interpreted as associations rather than causal effects.

2 Key words

Opioid Use Disorder, Regression Analysis, Public Health, Substance Use Disorder

3 Introduction

Opioid addiction is a public health crisis that has affected countless lives. The United States is experiencing a public health crisis of almost unprecedented scale: an epidemic of opioid use disorder (OUD) and related overdose deaths, the magnitude of which has increased exponentially in the past decades (Mancher & Leshner, 2019). In just one year, the age-adjusted rate of drug overdose deaths increased 31% (from 21.6 in 2019 per 100k population to 28.3 in 2020) (Hedegaard et al., 2021). In this project, we investigate what factors are important for predicting the prevalence rate of OUD across US states.

4 Literature review

OUD is a complex problem with many risk factors, including sociodemographic, geographic, and community/societal factors (Jalali et al., 2020). Sociodemographic risk factors include age, sex, and occupation. Opioid misuse is most common in young adults (18-25 years old), and first misuse of opioids at an early age is a significant risk factor for OUD (Crane, 2013; Schepis & Hakes, 2017). In addition, male sex is a risk factor for OUD because men are more likely than women to experience euphoria from opioids (Comer et al., 2010). Some occupations, particularly those that involve heavy physical labor (e.g., construction, mining), are also risk factors for OUD, because they may be associated with more work-related injuries that lead to chronic pain and treatment with opioid analgesics (Friedman et al., 2025; Shaw et al., 2020). In addition to these sociodemographic factors, mental health conditions, such as anxiety and depression, and a history of misusing other substances (e.g., illicit drugs) can increase the risk of OUD (Cicero et al., 2020; Valente et al., 2020).

The risk of OUD also varies by geographic location. For example, opioid medications are prescribed more frequently in non-metropolitan areas, which may be due to the higher proportion of older individuals in these areas who need pain relief after surgeries or individuals who are injured while performing manual labor (Keyes et al., 2014; Rigg & Monnat, 2015). Rural areas may also have fewer community resources for prevention and treatment programs than urban areas, which can increase the likelihood of developing and prolonging OUD (*Preventing Prescription Drug Misuse*, 2016).

Furthermore, disparities in healthcare access and insurance coverage can make it more difficult for some individuals to receive care for OUD, particularly those who are racial or ethnic minorities, live in rural areas, or cannot afford health insurance (Latimore et al., 2021; *Opioid Use Disorder*, 2019). The US Affordable Care Act (ACA; also known as “Obamacare”) provided states the option to extend Medicaid benefits to non-elderly adults with lower incomes, including access to OUD treatment; however, not all states have implemented the expansion (Mitchell, 2021). These differences in the implementation of ACA have resulted in a lower rate of prescriptions for medications to treat OUD in states that have not yet expanded Medicaid compared with those that have (Lewis et al.,

2020; Sharp et al., 2018).

Although many studies have documented risk factors for OUD in various groups (e.g., military veterans) or geographic regions (e.g., urban versus rural), few have analyzed nationwide data. One study, which modeled National Survey on Drug Use and Health (NSDUH) data from 2018-2019 + 2021-2024 using multivariate logistic regression, found that the most significant predictors of OUD were abuse of other substances, particularly tranquilizers (odds ratio [OR]=16.7), criminal history (OR=2.6), and lack of private health insurance (OR=2.1) (all $P < 0.001$) (Montiel Ishino et al., 2020). Here, we perform a similar analysis using a broader range of NSDUH data (2018-2019 + 2021-2024), including data before and after the Covid pandemic.

5 Methods

We use data on the state-level prevalence of pain reliever misuse and demographic characteristics from the National Survey on Drug Use and Health (NSDUH) from the Substance and Mental Health Services Administration for 2018-2019 and 2021-2024 (Substance Abuse and Mental Health Services Administration (SAMHSA)’s Data Analysis System (DAS), 2018-2024).

5.1 Exploratory Data Analysis

We performed exploratory data analysis to understand the distribution of our outcome variable (prevalence of pain reliever misuse) and the relationships between our predictors and the outcome. We visualized the correlations between predictors and the outcome variable, as well as the distribution of the numeric variables.

5.2 Data Preparation

Box-Cox transformations were used to normalize data distributions of numeric predictors. We confirmed that these variables had positive values, which is required for this method. We explicitly calculated the optimal lambda values, which provided insight into the optimal transformations needed to normalize each variable, and then used these values to calculate the transformed data.

The data did not contain any missing values, so imputation was not needed. For consistency with a similar study (Montiel Ishino et al., 2020) (see Literature review section), which will be used for comparison in the Discussion section, no additional features were created.

Finally, we split the data 70:30 into training and test sets.

5.3 Assumptions

Both models rely on standard OLS assumptions. We assumed a linear relationship between predictors and the outcome, assessed via residual-versus-fitted plots. Observations are independent, as each represents a distinct US state. Homoscedasticity was evaluated through residual plots, with

Box-Cox transformations applied to predictors to help stabilize variance. Normality of residuals was assessed using Q-Q plots. Multicollinearity was examined using variance inflation factors (VIFs); the stepwise AIC procedure’s variable selection partially addresses this by dropping redundant predictors.

LASSO uses the same basic assumptions of linear regression (linearity and independence), but it does not require residuals to be normally distributed. It works by adding a penalty that shrinks some coefficients to zero, which automatically removes less important variables and helps with multicollinearity. This is especially useful because the dataset has many correlated predictors compared to the number of observations (state-year data). The tuning parameter () was chosen using 10-fold cross-validation, and all predictors were standardized so they were on the same scale before fitting the model.

5.4 Model Building

For our models, we will attempt to predict the state-level prevalence of pain reliever misuse by evaluating a series of regression techniques designed to balance model fit with predictive accuracy. Our methodology follows a multi-stage approach, transitioning from a comprehensive baseline to restricted models that handle the complexities of high-dimensional demographic data.

5.4.1 Model Diagnostics

Model diagnostics were conducted to evaluate overall model adequacy and detect potential violations of assumptions. For both OLS-based models, diagnostic checks included residual-versus-fitted plots to assess linearity and homoscedasticity, Q-Q plots to evaluate normality of residuals, and leverage/Cook’s distance diagnostics to identify influential observations. No single state exhibited undue influence on model estimates, and residual patterns did not indicate strong systematic violations. These diagnostics supported the validity of the model specifications while also highlighting the expected limitations of applying linear models to aggregated state-level data.

5.4.2 Baseline Full Multiple Linear Regression

Our initial approach involves constructing a Full Multiple Linear Regression model using all 20+ available predictors. This process partitions the response variable into a systematic component represented by the predictors and a random error component. We use the method of Least Squares to calculate the unique set of coefficients that minimizes the sum of squared residuals, ensuring we obtain the best linear unbiased estimates (BLUE) given our data. To execute this, we will fit a model using every numeric demographic and socioeconomic variable in the training set to establish a performance ceiling for our R^2 and residual standard error.

5.4.3 Stepwise Variable Selection

To address potential redundancy in our 20+ predictors, we implement Stepwise Selection based on the Akaike Information Criterion (AIC). This “auto-selection” method iteratively adds or removes variables, comparing a larger model (Ω) to a smaller subset (ω). The goal is to minimize the AIC, which effectively penalizes the model for every additional parameter included, thereby favoring simpler, more interpretable explanations.

5.4.4 LASSO Regression

Because demographic variables (e.g., age, race, and poverty levels) are often highly correlated, we will utilize LASSO (Least Absolute Shrinkage and Selection Operator). This technique introduces a penalty term to the OLS equation that can shrink insignificant coefficients exactly to zero. This process requires Hyperparameter Tuning to find the optimal value of λ (the penalty strength) that produces the minimum error.

5.4.5 Predictive Validation and Testing

To ensure our models generalize to unseen data, we utilize a two-stage validation process. First, we will implement 10-fold Cross-Validation on the training data. This involves splitting the training set into ten equal parts, or “folds”; the model is trained on nine folds and validated on the tenth, with the process repeating ten times so every data point is used for validation once. We use the average Root Mean Squared Error (RMSE) across these folds to select our winning model. Finally, we will test the winning model on the held-out test set to get an honest estimate of its predictive ability on new data.

5.5 Model Selection

To determine the most effective predictive model for estimating state-level prevalence of pain reliever misuse, we compared three regression approaches: a Full Multiple Linear Regression model, a Stepwise AIC regression model, and a LASSO regression model. The purpose of comparing these models was to identify the model that provides the best balance between predictive accuracy, interpretability, and model simplicity.

The Full Multiple Linear Regression model was used as the baseline model because it included all transformed demographic and socioeconomic predictors. The Stepwise AIC model was used to reduce unnecessary predictors and create a more parsimonious model. The LASSO regression model was used because it can handle multicollinearity by shrinking less useful coefficients toward zero and performing variable selection.

All three models were evaluated using 10-fold Cross-Validation on the training dataset. Root Mean Squared Error (RMSE) was used as the main comparison metric because it measures prediction

error, with lower values indicating better predictive performance. The final model was selected based on cross-validated RMSE, model stability, and interpretability.

6 Results

```
# Read in all CSVs from the data directory and store them in a list
filepaths <- list.files(path = "data", pattern = "*.csv", full.names = TRUE)

# Helper to extract the "YY-YY" year substring from filenames
extract_year_tag <- function(path) {
  fname <- basename(path)
  tag <- stringr::str_extract(fname, "\\d{2}-\\d{2}")
  if (is.na(tag)) tag <- gsub("\\.csv$", "", fname)
  tag
}

# Read each file and add a 'year' column based on the filename (e.g. "18-19")
data_list <- lapply(filepaths, function(fp) {
  df <- readr::read_csv(fp, show_col_types = FALSE) |>
    janitor::clean_names()

  # Get name of first column (variable of interest)
  first_col <- names(df)[1]

  # Transform each df.
  df <- df |>
    # Filter out redundant ("Overall") rows
    filter(
      state_us_abbreviation != "Overall") |>

  # Calculate precise percentages
  group_by(state_us_abbreviation) |>
  mutate(
    denominator = weighted_count[column_percent == 1]
  ) |>
  ungroup() |>
  mutate(
    percent = weighted_count / denominator
  ) |>
```

```
# Get subset of columns
select(1, state_us_abbreviation, percent) |>

# Pivot first column wider
mutate(
  "{first_col}" := paste0(first_col, "_", as.character(.data[[first_col]]))
) |>
pivot_wider(names_from = 1, values_from = percent) |>

# Drop columns with all the same values
select(where(~ n_distinct(.) > 1)) |>

# Add year column
mutate(
  year = extract_year_tag(fp)
) |>

# Clean names
janitor::clean_names()

df
})

# Get list of data years
years <- sapply(filepaths, extract_year_tag) |> unique()

# Iterate across years, join data frames in data_list by state and year within each year.
yearly_joined_data <- lapply(years, function(yr) {
  year_dfs <- data_list |>
    purrr::keep(~ n_distinct(.x$year) == 1 && .x$year[[1]] == yr)

  if (length(year_dfs) == 0) return(NULL)
  if (length(year_dfs) == 1) return(year_dfs[[1]])

  Reduce(function(x, y) full_join(x, y, by = c("state_us_abbreviation", "year")), year_dfs)
})

# Append yearly_joined_data across all years into a single data frame
```

```
# Filter out NULL values first to avoid bind_rows errors
df <- yearly_joined_data |>
  purrr::discard(is.null) |>
  bind_rows()
```

```
df_renamed <- df |>
  rename(
    state = state_us_abbreviation,

    age_12_17 = rc_adult_age_category_recode_5_12_17_years_old,
    age_18_25 = rc_adult_age_category_recode_1_18_25_years_old,
    age_26_34 = rc_adult_age_category_recode_2_26_34_years_old,
    age_35_49 = rc_adult_age_category_recode_3_35_49_years_old,
    age_50_plus = rc_adult_age_category_recode_4_50_or_older,

    medicaid_coverage = covered_by_medicaid_chip_1_yes,

    less_than_high_school = rc_education_categories_1_less_high_school,
    high_school = rc_education_categories_2_high_school_grad,
    some_college = rc_education_categories_3_some_coll_assoc_dg,
    college_grad = rc_education_categories_4_college_graduate,

    emp_full_time = employment_status_imputation_revised_1_employed_full_time,
    emp_part_time = employment_status_imputation_revised_2_employed_part_time,
    emp_unemployed = employment_status_imputation_revised_3_unemployed,

    pain_reliefers_misuse = rc_pain_reliefers_past_year_misuse_1_misused_within_the_past_year,

    race_nh_white = rc_race_recode_4_levels_1_white_not_hispanic,
    race_nh_black = rc_race_recode_4_levels_2_black_not_hispanic,
    race_hisp = rc_race_recode_4_levels_4_hispanic,
    race_other = rc_race_recode_4_levels_3_other_or_multiple_not_hispanic,

    poverty = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_1_living_in_poverty,
    poverty_2x = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_2_income_up_to_2x_poverty,
    poverty_2x_plus = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_3_income_up_to_2x_plus_poverty,
  ) |>

  select(
```

```

state,
year,
pain_relievers_misuse,
age_12_17, age_18_25, age_26_34, age_35_49, age_50_plus,
medicaid_coverage,
less_than_high_school, high_school, some_college, college_grad,
emp_full_time, emp_part_time, emp_unemployed,
race_nh_white, race_nh_black, race_hisp, race_other,
poverty, poverty_2x, poverty_2x_plus
)

```

6.1 Exploratory Data Analysis

The data set contains 204 state-year rows and 23 features, and is 100 percent complete. The data are all numeric, and the target variable is `pain_relievers_misuse`, which represents the percentage of individuals who misused pain relievers in the past year. The other variables represent various demographic and socioeconomic characteristics of people surveyed in the state and year. See Table 1 for descriptive characteristics of the data.

```

tbl_summ_stats <- df_renamed |>
  skim_without_charts() |>
  janitor::clean_names() |>
  as_tibble() |>
  rename(Variable = skim_variable,
    "Completeness rate" = complete_rate,
    "Mean" = numeric_mean,
    "SD" = numeric_sd,
    "P50" = numeric_p50,
    "Min" = numeric_p0,
    "P25" = numeric_p25,
    "P75" = numeric_p75,
    "Max" = numeric_p100) |>
  select(Variable, "Completeness rate", Mean, Min, P25, P50, P75, Max) |>
  mutate(across(where(is.numeric), ~ signif(., 2))) |>
  knitr::kable()

tbl_summ_stats

```

Table 1: Summary statistics for all variables in the dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
state	1	NA	NA	NA	NA	NA	NA
year	1	NA	NA	NA	NA	NA	NA
pain_relievers_misuse	1	0.033	0.0120	0.026	0.032	0.039	0.058
age_12_17	1	0.091	0.0540	0.085	0.091	0.096	0.120
age_18_25	1	0.120	0.0980	0.120	0.120	0.130	0.160
age_26_34	1	0.140	0.1200	0.130	0.140	0.140	0.250
age_35_49	1	0.220	0.2000	0.210	0.220	0.230	0.260
age_50_plus	1	0.430	0.3100	0.410	0.430	0.440	0.500
medicaid_coverage	1	0.190	0.0500	0.150	0.190	0.230	0.370
less_than_high_school	1	0.084	0.0460	0.068	0.079	0.097	0.160
high_school	1	0.250	0.1300	0.220	0.250	0.270	0.370
some_college	1	0.280	0.1500	0.270	0.280	0.300	0.380
college_grad	1	0.290	0.1500	0.260	0.290	0.330	0.590
emp_full_time	1	0.440	0.2700	0.420	0.440	0.460	0.570
emp_part_time	1	0.130	0.0790	0.120	0.130	0.140	0.190
emp_unemployed	1	0.042	0.0170	0.033	0.041	0.050	0.075
race_nh_white	1	0.680	0.2100	0.580	0.710	0.800	0.940
race_nh_black	1	0.110	0.0042	0.032	0.072	0.150	0.440
race_hisp	1	0.120	0.0150	0.051	0.092	0.140	0.480
race_other	1	0.089	0.0230	0.047	0.065	0.095	0.680
poverty	1	0.150	0.0710	0.110	0.140	0.170	0.270
poverty_2x	1	0.190	0.0900	0.170	0.190	0.220	0.270
poverty_2x_plus	1	0.660	0.5000	0.620	0.670	0.720	0.780

```

# Get correlation matrix
df_renamed_cor <- df_renamed |>
  select(-state, -year) |>
  cor(use = "pairwise.complete.obs")

# Suppress small correlations by setting them to NA so corrrplot leaves them blank
df_renamed_cor[abs(df_renamed_cor) < 0.05] <- NA

corrrplot(df_renamed_cor,
  method = "circle",
  type = "upper",
  tl.cex = 0.4,

```

```

insig = "blank",
na.label = " ",
addCoef.col = "black",
number.cex = 0.3,
tl.srt = 45)

most_corr_w_target <- df_renamed_cor |>
  as.data.frame() |>
  rownames_to_column(var = "Variable") |>
  filter(Variable != "pain_relievers_misuse") |>
  transmute(
    Variable,
    Correlation = pain_relievers_misuse,
    Abs_Correlation = abs(pain_relievers_misuse)
  ) |>
  arrange(desc(Abs_Correlation)) |>
  filter(Abs_Correlation > 0.05) |>
  pull(Variable)

```

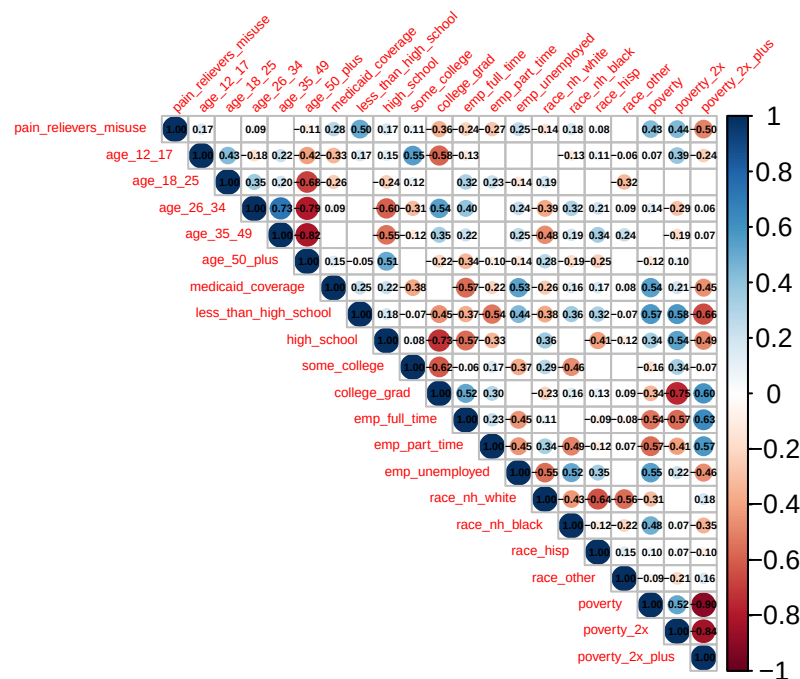


Figure 1: Correlation matrix of all non-ID variables in the dataset. Correlations < 0.05 are suppressed.

The correlation between all variables are shown in Figure 1. The following predictors are most

correlated with the target variable: less_than_high_school, poverty_2x_plus, poverty_2x, poverty, college_grad, medicaid_coverage, emp_part_time, emp_unemployed, emp_full_time, race_nh_black, high_school, age_12_17, race_nh_white, age_50_plus, some_college, age_26_34, race_hisp, which may be predictors of pain reliever misuse.

Lastly, we also looked at the distribution of each variable to check for skewness and outliers.

```
# Keep only continuous
continuous_vars <- df_renamed |>
  select(where(is.numeric)) |>
  names()

df_renamed |>
  select(all_of(continuous_vars)) |>
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  theme_minimal() +
  theme(panel.grid = element_blank()) +
  labs(x = NULL, y = "Count")
```

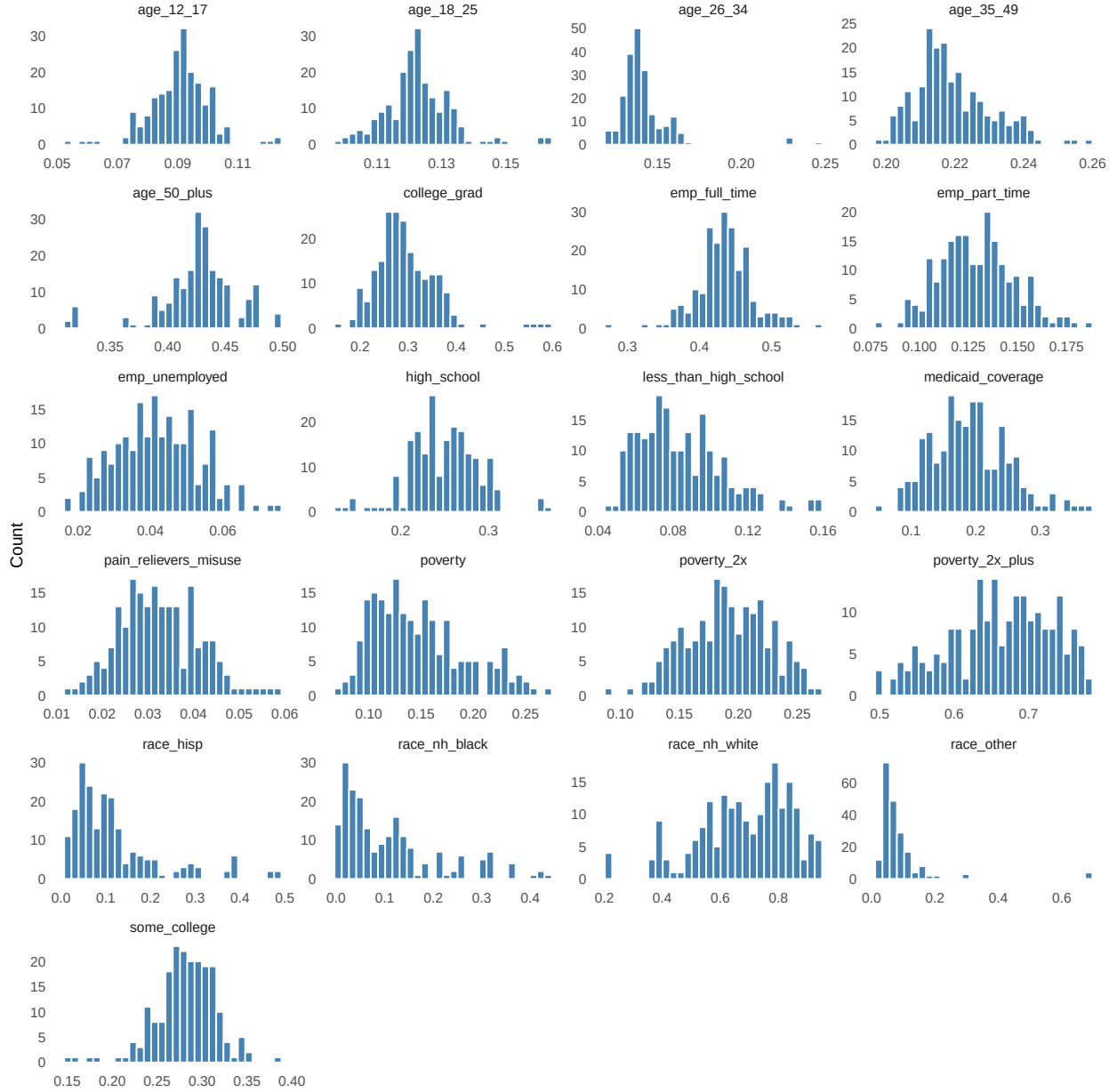


Figure 2: Distribution of numeric variables in the dataset.

Most variables are approximately normally distributed, except for many of the race indicator variables. The proportion of individuals who are non-white—especially those who are an “other” race besides hispanic, non-hispanic white, or non-hispanic black—are skewed right, mostly close to 0 across states and years, with a few higher outliers. This is expected given the racial demographics of the US. The proportion of individuals who are aged 26-34 also has a few high outliers.

6.2 Results of Data Preparation

6.2.1 Transformations

```
# Numeric predictors
predictors_untransformed <- select(df_renamed, age_12_17:poverty_2x_plus)

# Determine optimal lambda values for Box-Cox transformation
lambdas <- purrr::map_dbl(predictors_untransformed, ~ {
  caret::BoxCoxTrans(.x)$lambda
})

# Convert results vector to dataframe and format
optimal_lambdas <- as.data.frame(lambdas)
optimal_lambdas$variable <- rownames(optimal_lambdas)
rownames(optimal_lambdas) <- NULL
optimal_lambdas |>
  arrange(desc(lambdas)) |>
  select(variable, 'lambda' = 'lambdas') |>
  knitr::kable()
```

Table 2: Optimal lambda values for Box-Cox transformation

variable	lambda
age_50_plus	2.0
some_college	2.0
poverty_2x_plus	2.0
race_nh_white	1.9
age_12_17	1.4
emp_full_time	1.3
high_school	1.2
poverty_2x	1.2
emp_unemployed	0.7
medicaid_coverage	0.5
emp_part_time	0.3
race_nh_black	0.2
race_hisp	0.0
poverty	-0.2
less_than_high_school	-0.4
college_grad	-0.5

Table 2: Optimal lambda values for Box-Cox transformation

variable	lambda
race_other	-0.6
age_18_25	-1.1
age_26_34	-2.0
age_35_49	-2.0

Table 2 presents the optimal lambda values for Box-Cox transformation, which provided insight into which variables would benefit from various types of transformation. These lambda values were used to apply Box-Cox transformation to the numeric predictor variables.

- `age_50_plus`, `some_college`, `poverty_2x_plus`, and `race_nh_white` had $\lambda \approx 2$, which indicates squared transformation,
- Several variables (e.g., `age_12_17`, `poverty_2x`, `high_school`, `emp_full_time`) had $\lambda \approx 1$, which indicates that their distributions are approximately normal, so no (or very little) transformation is needed.
- `emp_unemployed` and `medicaid_coverage` had $\lambda \approx 0.5$, which indicates square root transformation,
- `race_nh_black`, `race_hisp`, and `poverty` had $\lambda \approx 0$, which indicates natural log transformation,
- `less_than_high_school`, `college_grad`, and `race_other` had $\lambda \approx -0.5$, which indicates reciprocal square root transformation,
- `age_18_25` had $\lambda \approx -1$, which indicates inverse transformation, and
- `age_26_34` and `age_35_49` had $\lambda = -2$, which indicates inverse squared transformation

```
# Box-Cox transformation of numeric variables using optimal lambda values (previous block)
lambdas_vec <- optimal_lambdas$lambdas
predictors_transformed <- purrr::map2_dfc(
  .x = predictors_untransformed,
  .y = lambdas_vec,
  .f = ~forecast::BoxCox(.x, .y)
)

# Restore the categorical and target variables and create new dataframe
df_prepared <- bind_cols(df_renamed[, 1:3], predictors_transformed)
```

```
# Removing the year and state variables since they are not needed for modeling and may cause d  
df_prepared <- df_prepared |> select(-state, -year)
```

6.2.2 Data split

Finally, we split the data 70:30 in preparation for model development.

```
set.seed(144)  
train_indices <- createDataPartition(df_prepared$pain_relievers_misuse, p = 0.7, list = FALSE)  
oud_train_split <- df_prepared[train_indices, ]  
oud_test_split <- df_prepared[-train_indices, ]
```

6.3 Assumptions

The Full Multiple Linear Regression and Stepwise AIC models rely on the standard assumptions of ordinary least squares (OLS) regression. These assumptions include linearity between predictors and the outcome, independence of observations, homoscedasticity (constant variance of residuals), approximate normality of residuals, and limited multicollinearity among predictors.

Because the dataset consists of state-year observations, we assume that observations are independent across states and survey years. Several predictors represent demographic proportions that sum to one within groups (e.g., race, age, and education categories), so some degree of multicollinearity was expected.

The LASSO model retains the assumptions of linearity and independence but addresses multicollinearity through L1 regularization, which shrinks less important coefficients toward zero. This makes LASSO particularly useful for datasets with many correlated predictors.

Diagnostic plots, variance inflation factors (VIFs), and residual tests were used after model fitting to evaluate whether these assumptions were reasonably satisfied.

6.4 Modeling

Before building the models, we define the 10-fold Cross-Validation logic that we will use for model selection and evaluation.

```
# Define 10-fold Cross-Validation logic  
# Ref: AMAR Ch 7.3 (Predictive Ability)  
train_control <- trainControl(method = "cv", number = 10)  
  
# List to contain modeling objects and cross-validation results  
model_results_list <- list()
```

Our First model is the Full Multiple Linear Regression model, which includes all predictors. This model serves as a baseline for comparison with more parsimonious models.

```
# 1. Full Multiple Linear Regression
model_full <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lm",
  trControl = train_control
)

model_full
```

Linear Regression

144 samples
20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 129, 129, 132, 130, 130, ...

Resampling results:

RMSE	Rsquared	MAE
0.008134775	0.2633035	0.006542457

Tuning parameter 'intercept' was held constant at a value of TRUE

```
model_results_list$Full_MLR <- model_full
```

Secondly, we will train our Stepwise Selection model, which iteratively adds or removes predictors based on AIC to find a more parsimonious model.

```
# 2. Stepwise Selection based on AIC
model_step <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lmStepAIC",
  trControl = train_control,
  trace = FALSE
)
```

```
model_step
```

Linear Regression with Stepwise Selection

144 samples

20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 130, 131, 130, 129, 129, ...

Resampling results:

RMSE	Rsquared	MAE
0.007938368	0.2375276	0.006570298

```
model_results_list$Stepwise_AIC <- model_step
```

Next, we train our LASSO regression model, which applies a penalty to shrink less important coefficients to zero, effectively performing variable selection. We implement a grid search to find the optimal value of the regularization parameter λ that minimizes the cross-validated error.

```
# 3. LASSO Regression with Hyperparameter Tuning
lasso_grid <- expand.grid(fraction = seq(0.1, 1, length = 10))

model_lasso <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lasso",
  trControl = train_control,
  tuneGrid = lasso_grid
)

model_lasso
```

The lasso

144 samples

20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 130, 129, 129, 129, 132, ...

Resampling results across tuning parameters:

fraction	RMSE	Rsquared	MAE
0.1	0.007728257	0.2588801	0.006212928
0.2	0.007867598	0.2504116	0.006315973
0.3	0.007929956	0.2487620	0.006404565
0.4	0.007921102	0.2525857	0.006410815
0.5	0.007916573	0.2561573	0.006407290
0.6	0.007906253	0.2611757	0.006384530
0.7	0.007896825	0.2667679	0.006364264
0.8	0.007910281	0.2683819	0.006379110
0.9	0.007929799	0.2698583	0.006391412
1.0	0.007964705	0.2702747	0.006409759

RMSE was used to select the optimal model using the smallest value.

The final value used for the model was fraction = 0.1.

```
model_results_list$LASSO <- model_lasso
```

Lastly, we will compare the performance of these three models based on their cross-validated RMSE and select the best performing model for final evaluation on the test set.

```
# Collecting CV results for comparison
cv_summary <- resamples(list(
  Full = model_results_list$Full_MLR,
  Stepwise = model_results_list$Stepwise_AIC,
  LASSO = model_results_list$LASSO
))

summary(cv_summary)
```

Call:

```
summary.resamples(object = cv_summary)
```

Models: Full, Stepwise, LASSO

Number of resamples: 10

MAE

Min.	1st Qu.	Median	Mean	3rd Qu.
------	---------	--------	------	---------

Full	0.005392517	0.006097335	0.006347721	0.006542457	0.006708942
Stepwise	0.004522811	0.005781265	0.006854365	0.006570298	0.007492335
LASSO	0.004355174	0.006085059	0.006320877	0.006212928	0.006711527

Max. NA's

Full	0.008036912	0
Stepwise	0.007695908	0
LASSO	0.007588355	0

RMSE

	Min.	1st Qu.	Median	Mean	3rd Qu.
Full	0.006458636	0.007418801	0.008002010	0.008134775	0.008819530
Stepwise	0.005656253	0.006872333	0.008239586	0.007938368	0.009044640
LASSO	0.005949474	0.007496975	0.007688121	0.007728257	0.008197995

Max. NA's

Full	0.010042840	0
Stepwise	0.009501905	0
LASSO	0.009275562	0

Rsquared

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
Full	0.01311226	0.1249150	0.2188552	0.2633035	0.3992077	0.5455291	0
Stepwise	0.01022618	0.1370386	0.2399172	0.2375276	0.3779030	0.4149258	0
LASSO	0.07537612	0.1787170	0.2877682	0.2588801	0.3219396	0.4198144	0

6.5 Model Diagnostics

```
# Extract final fitted lm objects from caret
lm_final   <- model_full$finalModel
step_final <- model_step$finalModel

# 2x4 grid: top row = Full MLR, bottom row = Stepwise AIC
par(mfrow = c(2, 4))
plot(lm_final, main = "Full MLR")
plot(step_final, main = "Stepwise AIC")
par(mfrow = c(1, 1))
```

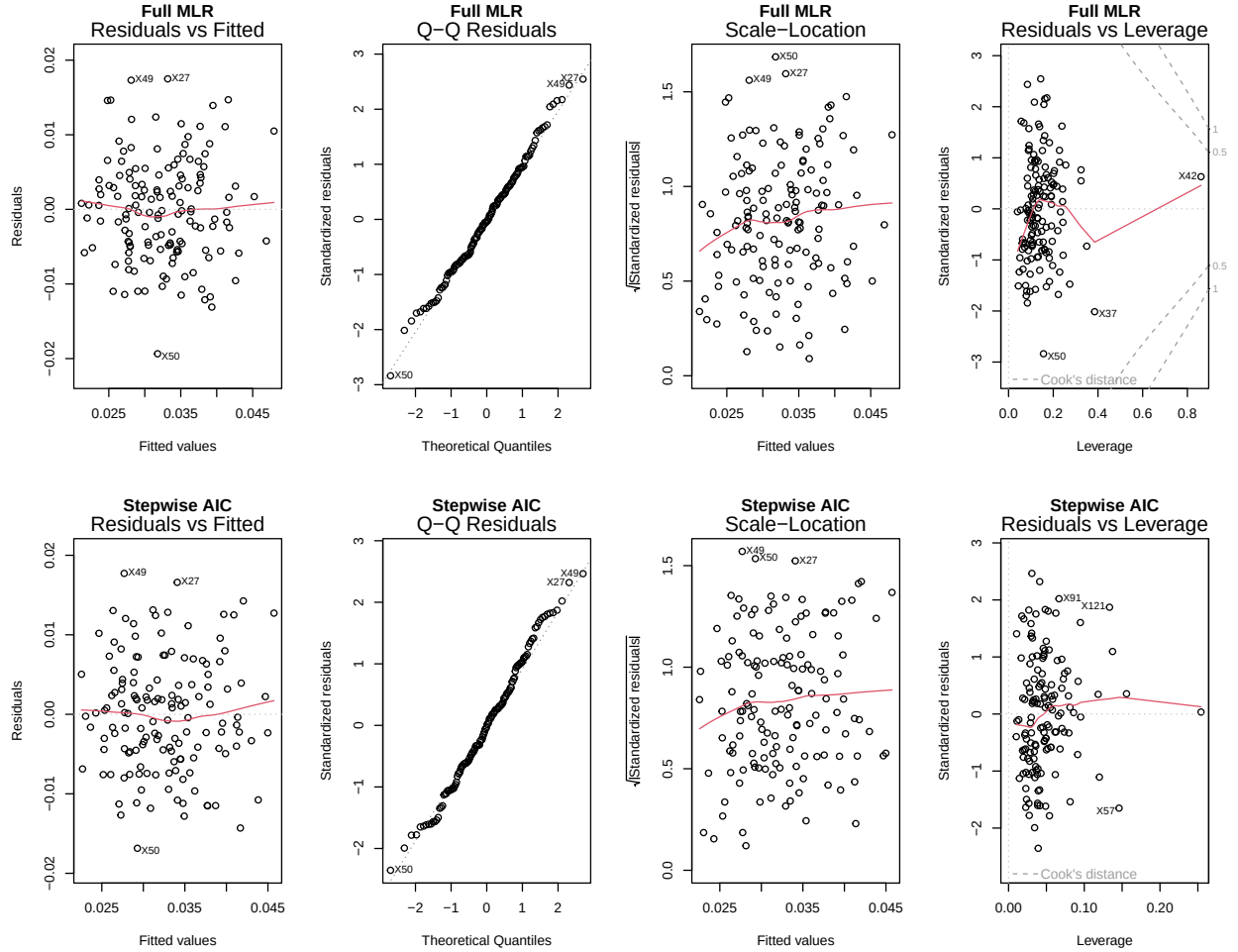


Figure 3: Diagnostic plots for the Full MLR and Stepwise AIC models

Linearity: Residual-versus-fitted plots for both models show no strong curvature and red lines approximately flat near zero, supporting the linearity assumption.

Normality: Q-Q plots track closely along the diagonal through the middle range, indicating approximate normality. Minor tail deviations are present in both models, with X50 as a consistent outlier, though this is expected given the small sample size.

Homoscedasticity: The Scale-Location plots show roughly stable variance across fitted values for both models, with a slight upward trend in the Full MLR that is reduced in the Stepwise AIC model.

Influential observations: X42 shows notably high leverage in the Full MLR, approaching Cook's distance boundaries. Leverage is more evenly distributed in the Stepwise AIC model, suggesting improved stability after variable selection.

```

vif_full <- car::vif(lm_final) |>
  as.data.frame() |>
  rownames_to_column("Predictor") |>
  rename(VIF = 2) |>
  arrange(desc(VIF)) |>
  mutate(VIF = round(VIF, 2))

knitr::kable(vif_full)

```

Table 3: Variance Inflation Factors — Full MLR

Predictor	VIF
age_50_plus	236.58
poverty_2x_plus	196.41
poverty	71.07
poverty_2x	46.41
age_26_34	39.08
age_35_49	39.08
college_grad	37.49
age_18_25	32.04
high_school	22.96
age_12_17	15.83
less_than_high_school	13.54
some_college	12.24
race_nh_white	10.34
race_other	5.39
race_hisp	4.83
emp_full_time	4.61
race_nh_black	3.94
medicaid_coverage	3.30
emp_part_time	3.24
emp_unemployed	3.10

VIF analysis showed strong multicollinearity among many predictors. **age_50_plus** (VIF = 236.6) and **poverty_2x_plus** (VIF = 196.4) had the highest values, while several age, poverty, education, and race variables exceeded the common threshold of 10. This was expected because many predictors are proportions that add up to 1 within categories such as age, race, and education, making them highly correlated with one another. In contrast, employment variables, **race_nh_black**, **race_hisp**, and **medicaid_coverage** had lower VIF values and showed little concern for multi-

collinearity. These results support the use of LASSO regression, which helps handle multicollinearity through regularization and variable selection.

Diagnostic plots for the Full MLR and Stepwise AIC models are shown in Figure 3. Overall, both models reasonably satisfied the main assumptions of linear regression. The residual plots showed no major nonlinear patterns, the Q-Q plots suggested approximate normality, and the Shapiro-Wilk tests indicated no strong evidence against normality (Full MLR $p = 0.655$, Stepwise AIC $p = 0.404$). The variance of residuals also appeared fairly consistent across fitted values. Observation X42 showed higher leverage in the Full MLR model, but this was reduced in the Stepwise AIC model.

VIF analysis (Table 3) showed strong multicollinearity among many predictors, especially `age_50_plus` (VIF = 236.6) and `poverty_2x_plus` (VIF = 196.4). This was expected because many predictors are demographic proportions that add up to 1 within categories such as age, race, and education. In contrast, employment variables, `race_nh_black`, `race_hisp`, and `medicaid_coverage` showed lower VIF values.

Because of this multicollinearity, LASSO regression was used as the final model. Its regularization penalty helps reduce instability from highly correlated predictors while also performing variable selection.

6.6 Results of model selection

```
# Collecting CV results for comparison
cv_summary <- resamples(list(
  Full = model_results_list$Full_MLR,
  Stepwise = model_results_list$Stepwise_AIC,
  LASSO = model_results_list$LASSO
))

summary(cv_summary)
```

Call:

```
summary.resamples(object = cv_summary)
```

Models: Full, Stepwise, LASSO

Number of resamples: 10

MAE

	Min.	1st Qu.	Median	Mean	3rd Qu.
Full	0.005392517	0.006097335	0.006347721	0.006542457	0.006708942

Stepwise	0.004522811	0.005781265	0.006854365	0.006570298	0.007492335
LASSO	0.004355174	0.006085059	0.006320877	0.006212928	0.006711527
	Max. NA's				
Full	0.008036912	0			
Stepwise	0.007695908	0			
LASSO	0.007588355	0			

RMSE

	Min.	1st Qu.	Median	Mean	3rd Qu.
Full	0.006458636	0.007418801	0.008002010	0.008134775	0.008819530
Stepwise	0.005656253	0.006872333	0.008239586	0.007938368	0.009044640
LASSO	0.005949474	0.007496975	0.007688121	0.007728257	0.008197995
	Max. NA's				
Full	0.010042840	0			
Stepwise	0.009501905	0			
LASSO	0.009275562	0			

Rsquared

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max. NA's
Full	0.01311226	0.1249150	0.2188552	0.2633035	0.3992077	0.5455291
Stepwise	0.01022618	0.1370386	0.2399172	0.2375276	0.3779030	0.4149258
LASSO	0.07537612	0.1787170	0.2877682	0.2588801	0.3219396	0.4198144

The three candidate models were compared using 10-fold Cross-Validation. The Full Multiple Linear Regression model provided a useful baseline because it included all available predictors, but it was also the most complex model and had a higher risk of overfitting.

The Stepwise AIC model improved model simplicity by removing predictors that contributed limited explanatory value. This made the model easier to interpret while still maintaining competitive predictive performance. The LASSO regression model provided the best overall balance between prediction accuracy and model simplicity. Because many demographic and socioeconomic predictors may be correlated, LASSO was useful because it reduced the effect of multicollinearity and selected the most important predictors through coefficient shrinkage.

Based on the cross-validated RMSE comparison, the LASSO regression model was selected as the final model for predicting state-level prevalence of pain reliever misuse. This model was preferred because it offered strong predictive performance, better generalizability, and a more stable structure for modeling public health data.

```
# 1. Extract coefficients from LASSO
# Caret's "lasso" uses the elasticnet package, so we use predict to get coefs at the best tuning
lasso_coefs <- predict(model_lasso$finalModel,
```

```

      s = model_lasso$bestTune$fraction,
      type = "coefficients",
      mode = "fraction")$coefficients

coef_lasso <- data.frame(
  Predictor = names(lasso_coefs),
  LASSO = as.numeric(lasso_coefs)
)

# Replace NA values with 0 (NAs occur when a model drops a predictor)
coef_lasso[is.na(coef_lasso)] <- 0

# 3. LASSO Importance (WORKAROUND: Absolute value of coefficients)
# Extract coefficients at the optimal lambda (same method as earlier)
lasso_coefs <- predict(model_lasso$finalModel,
  s = model_lasso$bestTune$fraction,
  type = "coefficients",
  mode = "fraction")$coefficients

imp_lasso <- data.frame(
  Predictor = names(lasso_coefs),
  LASSO = abs(as.numeric(lasso_coefs)) # Importance is absolute value of the coef
)

# Scale to 0-100
if(max(imp_lasso$LASSO, na.rm = TRUE) > 0) {
  imp_lasso$LASSO <- (imp_lasso$LASSO / max(imp_lasso$LASSO, na.rm = TRUE)) * 100
}

# Replace NA values with 0 (occurs when a model drops a predictor)
imp_lasso[is.na(imp_lasso)] <- 0

# Join coefficients and importance into one table, adding lambda values
imp_coefs_table <- coef_lasso |>
  dplyr::left_join(imp_lasso, by = "Predictor", suffix = c(" Coefficient", " Importance")) |>
  dplyr::mutate(
    dplyr::across(where(is.numeric), ~ round(., 3))
  ) |>
  dplyr::arrange(dplyr::desc(`LASSO Importance`))

```

```
# Print the resulting dataframe
knitr::kable(imp_coeffs_table)
# Get list of most important predictors
most_imp_predictors <- imp_coeffs_table |>
  filter(`LASSO Importance` > 0) |>
  pull(Predictor)
```

Table 4: Coefficients and Importance of the Lasso Model

Predictor	LASSO Coefficient	LASSO Importance
some_college	0.084	100.000
poverty_2x_plus	-0.029	34.583
age_12_17	0.012	14.430
emp_full_time	0.010	11.425
medicaid_coverage	0.007	8.143
poverty_2x	0.005	5.902
less_than_high_school	0.004	5.300
college_grad	-0.002	2.239
race_nh_white	-0.001	1.624
race_nh_black	-0.001	1.279
age_26_34	0.000	0.275
age_18_25	0.000	0.263
race_other	0.000	0.154
age_35_49	0.000	0.000
age_50_plus	0.000	0.000
high_school	0.000	0.000
emp_part_time	0.000	0.000
emp_unemployed	0.000	0.000
race_hisp	0.000	0.000
poverty	0.000	0.000

Table 4 shows the coefficients and importance of each predictor in our final LASSO regression model. The final LASSO model identified the following predictors as most important for predicting pain reliever misuse: some_college, poverty_2x_plus, age_12_17, emp_full_time, medicaid_coverage, poverty_2x, less_than_high_school, college_grad, race_nh_white, race_nh_black, age_26_34, age_18_25, race_other. These predictors had non-zero coefficients in the LASSO model, indicating that they were selected as significant contributors to the prediction of pain reliever misuse.

```
# Predict on the held-out test set using the final LASSO model
test_preds <- predict(model_lasso, newdata = oud_test_split)

# Calculate final performance metrics
test_rmse <- caret::RMSE(test_preds, oud_test_split$pain_relievers_misuse)
test_r2    <- caret::R2(test_preds, oud_test_split$pain_relievers_misuse)

# Display metrics
data.frame(Test_RMSE = test_rmse, Test_R2 = test_r2) |>
  knitr::kable(caption = "Final LASSO Model Performance on the Held-Out Test Set")
```

Table 5: Final LASSO Model Performance on the Held-Out Test Set

Test_RMSE	Test_R2
0.0057538	0.4907906

To assess its generalizability, the final LASSO model was evaluated on the held-out test set, yielding an RMSE of 0.006 and an R^2 of 0.491. These metrics confirm that the model retains strong predictive power and stable performance when applied to unseen data, showing no signs of significant overfitting compared to the training phase.

7 Discussion

Our results suggested that educational attainment was one of the strongest predictors of pain reliever misuse. The rate of folks having completed some college education was positively associated with the outcome ($\beta = 0.084$), while the proportions with less than a high school education ($\beta = 0.004$) and with a college degree ($\beta = -0.002$) were both negatively associated with the outcome. In other words, states with fewer educational resources or lower educational attainment tended to have higher rates of pain reliever misuse.

Income also mattered. The share of folks with incomes more than two times the federal poverty level (FPL) was negatively associated with the outcome ($\beta = -0.029$), while the share living at or below that threshold was positively associated with the outcome ($\beta = 0.005$). For 2026, 200% of the FPL in most states is approximately \$31,920 for a single person, \$43,280 for a 2-person household, \$54,640 for 3, and \$66,000 for a family of 4, annually (*Federal Poverty Level (FPL)*, 2026). This pattern suggests that states with more residents living closer to the poverty threshold tended to have higher rates of pain reliever misuse, while states with more residents above that level tended to have lower rates.

Age composition was also important. The proportion of children aged 12 to 17 was positively

associated with the outcome ($\beta = 0.012$), but adult age groups were not strongly predictive in the final model. This suggests that, states with a higher share of the youth population tended to have higher rates of OUD, and child age was more informative than variation across adult age groups.

Employment and insurance coverage were also associated with the outcome. Full-time employment was positively associated with pain reliever misuse ($\beta = 0.01$), and Medicaid coverage was also positively associated with the outcome ($\beta = 0.007$). These relationships may reflect differences in healthcare access, prescription exposure, and the underlying vulnerability of populations covered by Medicaid.

Race composition also appeared in the final model. The proportions of non-Hispanic white and non-Hispanic Black residents were both negatively associated with the outcome ($\beta = -0.001$ and $\beta = -0.001$, respectively). These coefficients should be interpreted as state-level associations rather than individual-level effects, and they may be capturing broader social and structural differences across states.

Overall, the final LASSO model suggests that educational attainment, poverty, age composition, employment, insurance coverage, and racial composition all help explain variation in state-level pain reliever misuse. However, these are associations rather than causal effects, and the coefficients should be read as model-based summaries of the transformed predictors.

The associations identified in our LASSO model were in agreement with risk factors identified in a previous study using a logistic regression model; however, there were some differences in the relative importance of predictors. (Montiel Ishino et al., 2020) These differences may reflect the longer study period in our analysis as well as differences in modeled predictors (i.e., Montiel-Ishino et al. included abuse of non-opioid substances).

Our study has several limitations. Namely, the data are at the state level, so we cannot make inferences about individual-level risk factors for OUD. In addition, our data are cross-sectional, so we cannot make causal inferences about the relationships between predictors and OUD. Finally, our data only include a limited number of predictors, so there may be other important factors that we did not include in our models.

8 Conclusions

We found that state-level pain reliever misuse was most strongly associated with educational attainment, poverty, age composition, employment, Medicaid coverage, and racial composition. In the final LASSO model, some college education, the share of children aged 12 to 17, full-time employment, and Medicaid coverage were positively associated with pain reliever misuse, while higher shares of people with a college degree, less than a high school education, incomes above two times the federal poverty level, and non-Hispanic white and Black populations were negatively associated with misuse.

These results should be interpreted as state-level associations rather than causal effects. Our analysis is also limited by the use of aggregated cross-sectional data and by the relatively small set of predictors available in NSDUH. Future research could use longitudinal data, include additional structural and behavioral predictors, and compare linear and penalized regression approaches with nonlinear or tree-based models.

References

- Cicero, T. J., Ellis, M. S., & Kasper, Z. A. (2020). Polysubstance use: A broader understanding of substance use during the opioid crisis. *Am. J. Public Health*, 110(2), 244–250. <https://doi.org/10.2105/AJPH.2019.305412>
- Comer, S. D., Cooper, Z. D., Kowalczyk, W. J., Sullivan, M. A., Evans, S. M., Bisaga, A. M., & Vosburg, S. K. (2010). Evaluation of potential sex differences in the subjective and analgesic effects of morphine in normal, healthy volunteers. *Psychopharmacology (Berl.)*, 208(1), 45–55. <https://doi.org/10.1007/s00213-009-1703-4>
- Crane, E. H. (2013). *Highlights of the 2011 drug abuse warning network (DAWN) findings on drug-related emergency department visits*. <https://www.ncbi.nlm.nih.gov/books/NBK384680/>
- Federal poverty level (FPL). (2026). HealthCare.gov; <https://www.healthcare.gov/glossary/federal-poverty-level-fpl/>.
- Friedman, L. S., Go, L. H. T., Dang, N., Shannon, B., Bonney, T., Richardson, D., Cohen, R. A., & Almberg, K. S. (2025). The association between employment in coal mining and history of injury, current pain, and prescription opioid use. *Am. J. Ind. Med.*, 68(1), 76–88. <https://doi.org/10.1002/ajim.23679>
- Hedegaard, H., Miniño, A. M., Spencer, M. R., & Warner, M. (2021). *Drug overdose deaths in the united states, 1999–2020* (No. 428). <http://dx.doi.org/10.15620/cdc:112340>
- Jalali, M. S., Botticelli, M., Hwang, R. C., Koh, H. K., & McHugh, R. K. (2020). The opioid crisis: A contextual, social-ecological framework. *Health Res. Policy Syst.*, 18(1), 87. <https://doi.org/10.1186/s12961-020-00596-8>
- Keyes, K. M., Cerdá, M., Brady, J. E., Havens, J. R., & Galea, S. (2014). Understanding the rural-urban differences in nonmedical prescription opioid use and abuse in the united states. *Am. J. Public Health*, 104(2), e52–9. <https://doi.org/10.2105/AJPH.2013.301709>
- Latimore, A., Hinojosa, A. O., Kestner, L., & Patel, P. (2021). *Exploring urban-rural disparities in accessing treatment for opioid use disorder*. American Institutes for Research. <https://www.air.org/resource/field/exploring-urban-rural-disparities-accessing-treatment-opioid-use-disorder>
- Lewis, R., Baugher, A. R., Finlayson, T., Wejnert, C., Sionean, C., & National HIV Behavioral Surveillance (NHBS) Study Group. (2020). Healthcare access and utilization among persons who inject drugs in medicaid expansion and nonexpansion states: 22 united states cities, 2018. *J. Infect. Dis.*, 222(Suppl 5), S420–S428. <https://doi.org/10.1093/infdis/jiaa337>
- Mancher, M., & Leshner, A. I. (2019). *Medications for opioid use disorder save lives*. National

Academies Press.

- Mitchell, A. (2021). *Overview of the ACA medicaid expansion*. Congressional Research Service. <https://www.congress.gov/crs-product/IF10399>
- Montiel Ishino, F. A., McNab, P. R., Gilreath, T., Salmeron, B., & Williams, F. (2020). A comprehensive multivariate model of biopsychosocial factors associated with opioid misuse and use disorder in a 2017-2018 united states national survey. *BMC Public Health*, 20(1), 1740. <https://doi.org/10.1186/s12889-020-09856-2>
- Opioid use disorder: Challenges and opportunities in rural communities*. (2019). PEW Charitable Trusts. https://www.pew.org/-/media/assets/2019/02/opioiduseruralcommunities_final.pdf
- Preventing prescription drug misuse: Overview of factors and strategies*. (2016). EDC Prevention Solutions. https://solutions.edc.org/sites/default/files/Preventing-Prescription-Drug-Misuse-Overview-Factors-Strategies_0.pdf
- Rigg, K. K., & Monnat, S. M. (2015). Urban vs. Rural differences in prescription opioid misuse among adults in the united states: Informing region specific drug policies and interventions. *Int. J. Drug Policy*, 26(5), 484–491. <https://doi.org/10.1016/j.drugpo.2014.10.001>
- Schepis, T. S., & Hakes, J. K. (2017). Age of initiation, psychopathology, and other substance use are associated with time to use disorder diagnosis in persons using opioids nonmedically. *Subst. Abus.*, 38(4), 407–413. <https://doi.org/10.1080/08897077.2017.1356791>
- Sharp, A., Jones, A., Sherwood, J., Kutsa, O., Honermann, B., & Millett, G. (2018). Impact of medicaid expansion on access to opioid analgesic medications and medication-assisted treatment. *Am. J. Public Health*, 108(5), 642–648. <https://doi.org/10.2105/AJPH.2018.304338>
- Shaw, W. S., Roelofs, C., & Punnett, L. (2020). Work environment factors and prevention of opioid-related deaths. *Am. J. Public Health*, 110(8), 1235–1241. <https://doi.org/10.2105/AJPH.2020.305716>
- Substance Abuse and Mental Health Services Administration (SAMHSA)’s Data Analysis System (DAS). (2018-2024). *NSDUH: 2-year restricted-use data (2018-2019, 2021-2022, 2022-2023, 2023-2024)*. https://datatools.samhsa.gov/das/nsduh/2022/nsduh-2022-2023-rd02yr/crosstab?row=RACE4&column=STUSAB&weight=DASWT_1
- Valente, P. K., Bazzi, A. R., Childs, E., Salhaney, P., Earlywine, J., Olson, J., Biancarelli, D. L., Marshall, B. D. L., & Biello, K. B. (2020). Patterns, contexts, and motivations for polysubstance use among people who inject drugs in non-urban settings in the U.S. northeast. *Int. J. Drug Policy*, 85(102934), 102934. <https://doi.org/10.1016/j.drugpo.2020.102934>